MACHINE LEARNING ASSIGNMENT - 1

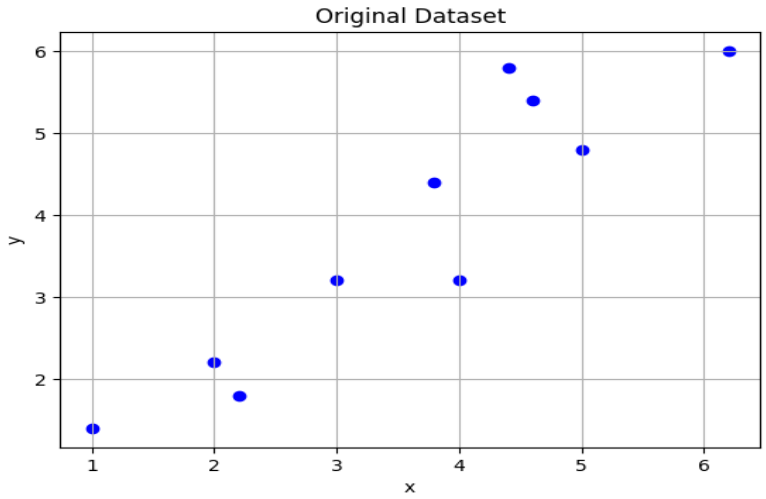
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1. Introduction

Principal Component Analysis (PCA) is a dimensionality reduction technique used to identify the principal components in a dataset. This report documents the steps taken to perform PCA on the given dataset, including visualization, normalization, eigen decomposition, transformation, and reconstruction.

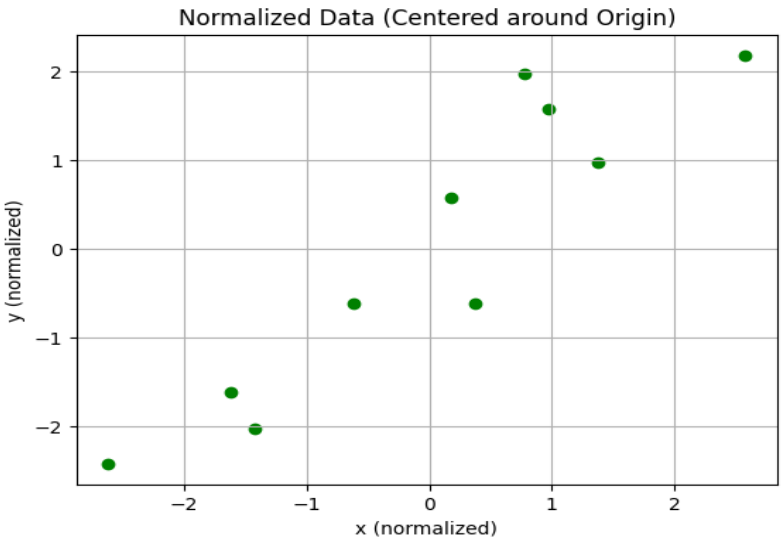
2. Dataset Description

The dataset consists of 10 observations with two characteristics (x, y). The original dataset was visualized using a scatter plot to understand its distribution.



3. Data Normalization

The data was normalized by centering it around the origin, achieved by subtracting the mean of each feature. The normalized dataset was then plotted to visualize the transformation.



4. Covariance Matrix Calculation

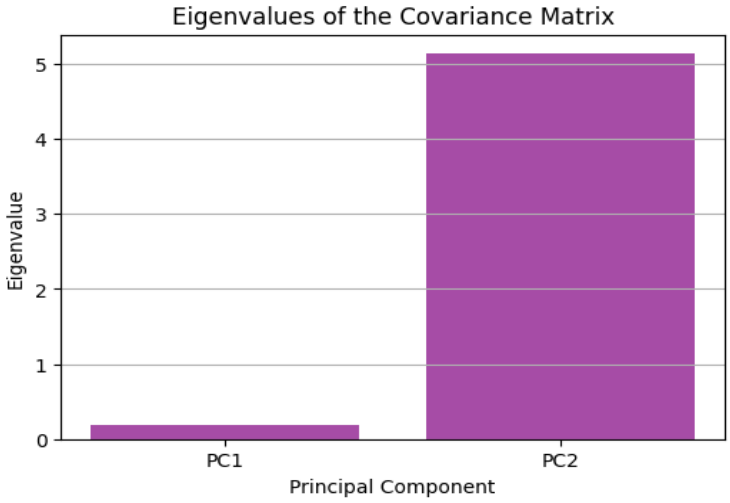
A covariance matrix was computed to analyze the relationships between features. This matrix represents how the features vary together.
Covariance Matrix:
[[2.46622222 2.46177778] [2.46177778 2.86622222]]

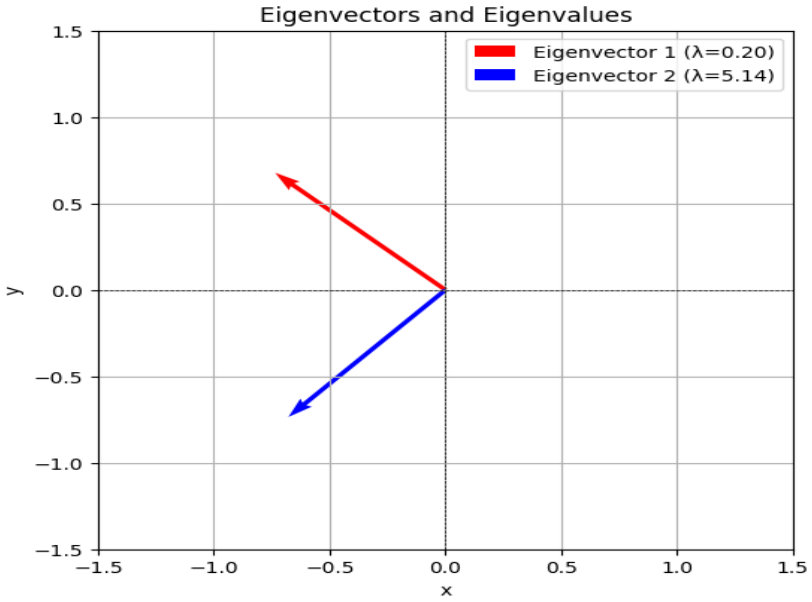
5. Eigenvalues and Eigenvectors

The eigenvalues and eigenvectors of the covariance matrix were calculated to determine the principal components. The computed eigenvalues and eigenvectors are as follows:

Eigenvalue	Eigenvector X	Eigenvector Y
0.1963336	-0.735179	0.677873
5.1361108	-0.677873	-0.735179

The largest eigenvalue (5.1361108) corresponds to the principal component (-0.677873, -0.735179), which explains the most variance in the dataset.





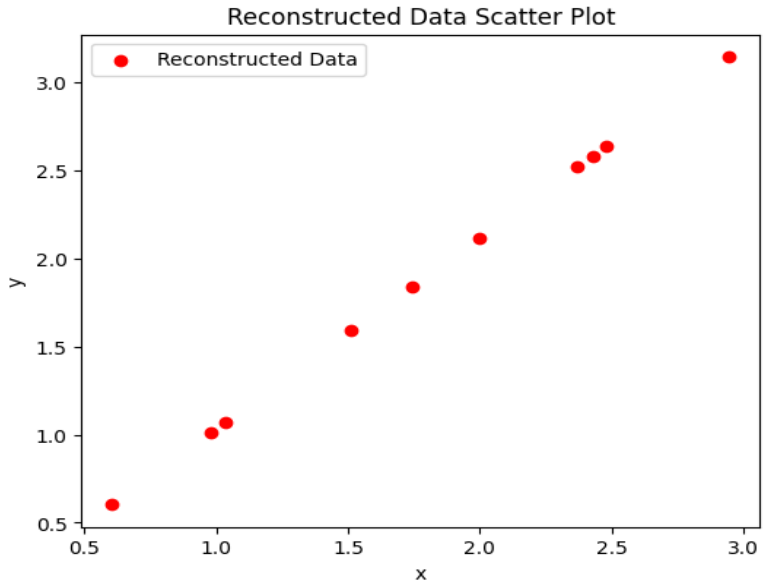
6. Transformation to Lower Dimension

The dataset was projected onto the principal component to reduce its dimensionality while preserving the most significant variance.
[-1.65594037 3.55516065 -1.98439499 -0.54842083 -3.35160284 -1.82589821

 0.19821887 2.28914433 0.87609227 2.44764111]

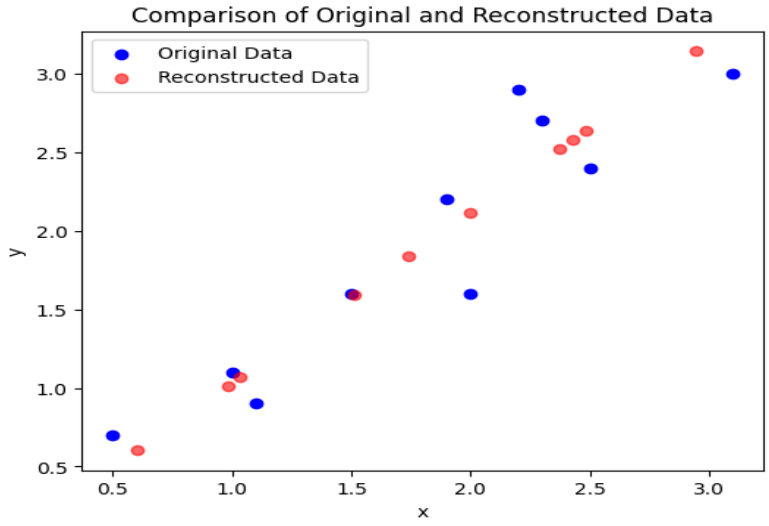
7. Data Reconstruction

The transformed data was projected back onto the original space using the principal component and adding back the mean to reconstruct an approximation of the original dataset.



8. Visualization and Comparison

- The original dataset was plotted alongside the reconstructed data.
- The reconstructed data closely follows the original data, indicating that the transformation and reconstruction were effective in preserving the dataset's structure.



9. Conclusion

PCA successfully reduced the dataset's dimensionality while retaining its core characteristics. The reconstruction process demonstrated that most of the dataset's information was preserved using just one principal component. This technique is useful in high-dimensional datasets where reducing dimensions can enhance efficiency while maintaining meaningful insights.